

High-energy neutrino astrophysics

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The chargeless, weakly interacting neutrinos are ideal astronomical messengers as they travel through space without scattering, absorption or deflection. But this weak interaction also makes them notoriously difficult to detect, leading to neutrino observatories requiring large-scale detectors. A few years ago, the IceCube experiment discovered neutrinos originating beyond the Sun with energies bracketed by those of the highest energy gamma rays and cosmic rays. I discuss how these high-energy neutrinos can be detected and what they can tell us about the origins of cosmic rays and about dark matter.

The idea that the neutrino is an ideal astronomical messenger^{1–3} emerged soon after the particle was first detected⁴. Neutrinos can reach us from the edge of the universe without being absorbed or deflected by magnetic fields. They can escape unscathed from the inner neighbourhood of black holes and from the accelerators where cosmic rays are born.

Cosmic rays—extremely high-energy protons and heavier nuclei—constantly bombard Earth from space with energies exceeding a million times those of present accelerator beams. More than a century after their discovery, the processes that generate them are still unknown. And we cannot pinpoint where in the universe cosmic rays come from because their electric charge allows galactic and intergalactic magnetic fields to alter their course. Luckily, however, theory suggests that at their birthplaces cosmic rays also interact with radiation fields and matter to produce neutrinos. Although cosmic rays themselves cannot lead us to their origins, the highly energetic neutrinos they presumably produce may do so. The possibility of observing the so-called Greisen–Zatsepin–Kuzmin (GZK) neutrinos produced in the interactions of cosmic rays with microwave background photons⁵ was recognized by 1969⁶. Decades of theoretical modelling led to the conclusion that kilometre-long detectors would be required to collect neutrinos from the cosmos in the statistically significant numbers^{7–10} necessary to finally identify the sources of the cosmic rays.

At the end of its life, a massive star may collapse into a neutron star or a black hole. If a few per cent of the released gravitational energy is converted into thrust for the acceleration of particles, presumably through shock waves, one can accommodate the observed energy spectra of cosmic rays. This conversion can take thousands of years in a supernova remnant¹¹ or seconds in a gamma-ray burst¹² that, theoretically, can produce the highest energy extragalactic cosmic rays with energies exceeding 10^{20} eV. A consensus emerged that such sources could release a similar energy in cosmic rays, gamma rays, and neutrinos¹³. This predicted neutrino flux was in fact discovered in 2013 by the IceCube experiment (see Box 1).

Following the cues from astronomical telescopes, IceCube has been searching for neutrinos arriving from the direction and at the time of a gamma-ray burst. After more than one thousand follow-up observations, none was found, resulting in a limit on the neutrino flux from gamma-ray bursts of less than one per cent. This has shifted the focus to an alternative explanation for the sources of extragalactic cosmic rays: active galactic nuclei. This is a class of

galaxies powered by supermassive black holes at their centres, which accelerate cosmic rays in shocks in the accretion disk or the jets associated with the black hole.

Even at a depth of 1,450 m, IceCube detects at a rate of 3,000 per second a background of atmospheric cosmic-ray muons originating in the Southern Hemisphere. Two methods are used to identify neutrinos. Historically, neutrino searches focus on the observation of muon neutrinos that interact primarily outside the detector to produce kilometre-long muon tracks passing through the detection volume. Although this allows for the observation of neutrinos that interact outside the detector, it is then necessary to use the Earth as a filter remove the huge background of cosmic-ray muons. This limits the neutrino view to a single flavour and half the sky. The alternative method exclusively identifies neutrinos interacting inside the detector. It divides the instrumented volume of ice into an outer veto shield and a ~ 500 megaton inner fiducial volume. The advantage of focusing on neutrinos interacting inside the instrumented volume of ice is that the detector functions as a total absorption calorimeter measuring energy with a 10–15% resolution. Also, neutrinos from all directions in the sky can be identified, including both muon tracks, produced in muon neutrino charged-current interactions, and secondary showers, produced by electron and tau neutrinos as well as in neutral current interactions of neutrinos of all flavours. The Čerenkov patterns initiated by an electron (or tau) neutrino of 1 PeV (petaelectronvolt, or 10^{15} eV) energy and a muon neutrino depositing 2.6 PeV energy while traversing the detector are contrasted in Fig. 1.

In general, the particle's trajectory is determined from the arrival times of photons at the optical sensors¹⁴, whereas the number of photons is a proxy for the deposited energy. The two methods for separating neutrinos from the cosmic-ray muon background have complementary advantages. The long tracks produced by muon neutrinos can be pointed back to their sources with a 0.3° angular resolution. In contrast, the reconstruction of the direction of secondary showers, still in the development stage in IceCube¹⁵, can be pointed to within 10° – 15° of the direction of the incident neutrino. Determining the deposited energy from the observed light pool is, however, relatively straightforward, and a resolution of better than 15% is possible.

For neutrino astronomy, the first challenge is to select sufficiently pure samples of neutrinos: roughly 100,000 per year in a background of ten billion cosmic-ray muons. The second is to identify the small fraction of neutrinos that are those of astrophysical origin, expected

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Box 1 | The IceCube experiment.

Given the large size of neutrino detectors, early efforts concentrated on using large volumes of natural water and photomultipliers^{50,51} to detect the Čerenkov light emitted by secondary particles produced when neutrinos interact with nuclei inside or near the detector^{52–57}. In contrast, the IceCube detector^{58–60} transforms deep natural Antarctic ice 1,450 m below the geographic South Pole into a Čerenkov detector. The instrument consists of 5,160 digital optical modules that instrument a cubic kilometre of ice. Each digital optical module

consists of a glass sphere that contains a 10-inch photomultiplier and the electronics board that digitizes the signals locally using an onboard computer. The digitized signals are given a global time stamp with residuals accurate to 2 ns and are subsequently transmitted to the surface. Processors at the surface continuously collect the time-stamped signals from the optical modules, each of which functions independently. These signals are sorted into telltale patterns of light that reveal the direction, energy, and flavour of the incident neutrino.

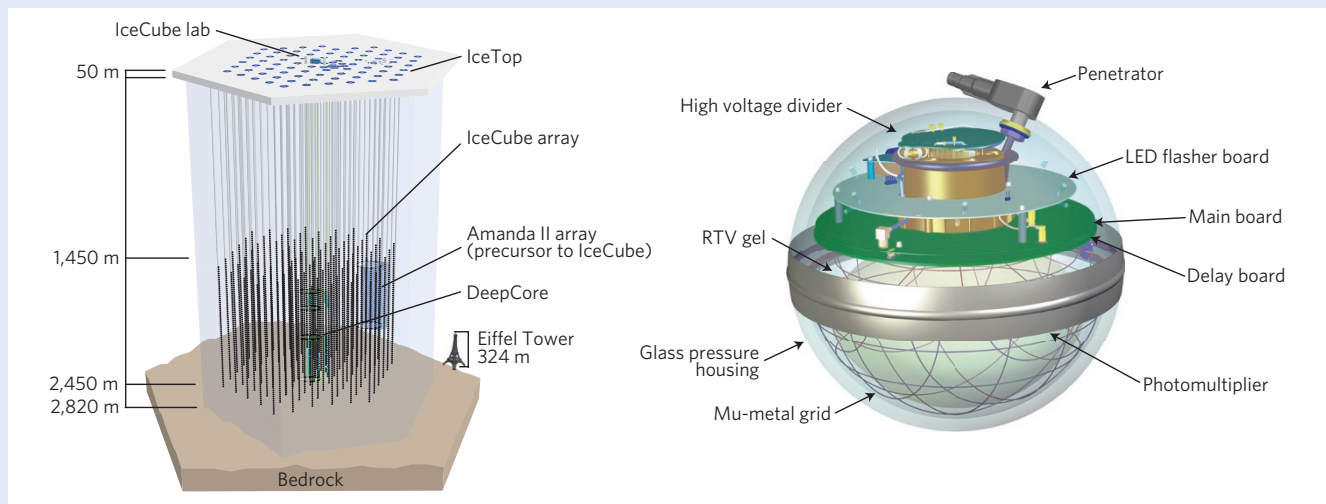


Figure B1 | Sketch of the IceCube observatory (left) and the digital optical module (right). RTV, room temperature vulcanization.

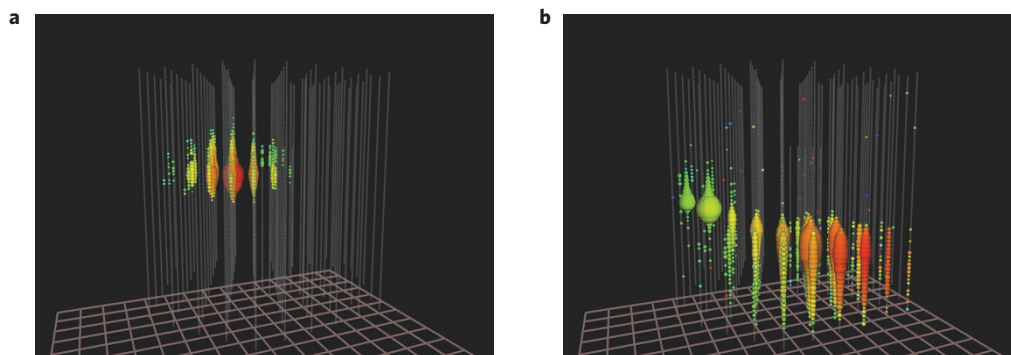


Figure 1 | Light pools produced in IceCube by showers initiated by an electron or a tau neutrino, or the neutral current interaction of a neutrino of any of the three flavours. For the coloured dots, the colour indicates arrival time, from red (early) to purple (late) following the rainbow, and size reflects the number of photons detected. **a**, The measured energy of the shower is 1.14 PeV, which represents a lower limit on the energy of the neutrino that initiated the shower. White dots represent sensors with no signal. **b**, An upgoing muon track traverses the detector at an angle of 11° below the horizon. The deposited energy inside the detector is 2.6 PeV.

at the level of tens of events per year. Atmospheric neutrinos are an overwhelming background for cosmic neutrinos, at least at energies below ~ 100 TeV. Above this energy, however, the atmospheric neutrino flux is too small to produce events, even in a kilometre-scale detector, and every event in that range is a discovery.

By now, IceCube has observed cosmic neutrinos using both methods for rejecting background; each analysis has reached a statistical significance of more than 6σ (refs 16,17). Based on different methods for reconstruction and energy measurement, their results agree, pointing at extragalactic sources whose flux has equilibrated in the three flavours after propagation over cosmic

distances. Its total energy matches that of extragalactic photons and cosmic rays.

Detection of cosmic neutrinos

IceCube is sensitive to neutrinos with energies above a threshold of approximately 0.1 TeV. Using the Earth as a filter, a flux of neutrinos is identified that is consistent with atmospheric origin; see Fig. 2. However, in seven years of data, a clear excess of events (6σ significance) is observed at energies beyond 100 TeV (refs 17–19), which cannot be accommodated by the atmospheric flux. Whereas IceCube measures only the energy deposited by the

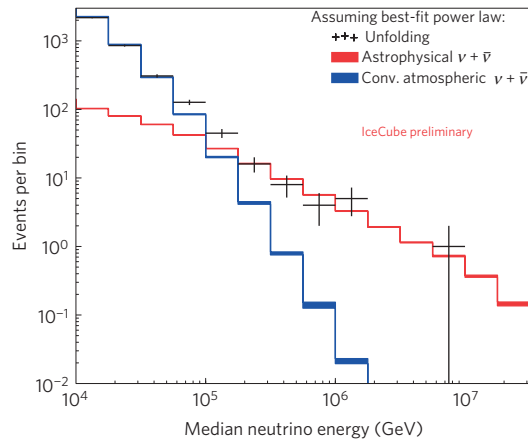


Figure 2 | Spectrum of secondary muons initiated by muon neutrinos that have traversed the Earth, that is, with zenith angle less than 5° above the horizon, as a function of the energy they deposit inside the detector. For each reconstructed muon energy, the median neutrino energy is calculated assuming the best-fit spectrum. The coloured bands (blue/red) show the expectation for the conventional and astrophysical contributions. The black crosses show the measured data with the vertical bars indicating measurement error.

secondary muon inside the detector, Standard Model physics allows one to infer the energy spectrum of the parent neutrinos shown in Fig. 2. For the highest energy event, shown in Fig. 1, the most likely energy of the parent neutrino is almost 10 PeV. The muon energy loss measured in the detector is 2.6 ± 0.3 PeV. The cosmic flux is well described by a power law with a spectral index of 2.13 ± 0.13 and a normalization at 100 TeV neutrino energy of $(0.9 \pm 0.3) \times 10^{-18} \text{ GeV}^{-1} \text{ cm}^{-2} \text{ sr}^{-1}$. The neutrino energy contributing to this flux covers the energy range of 200 TeV to 9 PeV. We find, however, that other analyses that extend to lower energies indicate an excess of neutrino interaction events from all sources in the 30–100 TeV energy range, well above the extrapolation of this cosmic neutrino flux measurement to lower energies. The conclusion to be drawn is that the astrophysical flux measured by IceCube is not featureless.

The first hint for cosmic neutrinos emerged when two events were found serendipitously in a search for GZK neutrinos using IceCube data collected between May 2010 and May 2012²⁰. The energies of these neutrinos, rather than EeV (exaelectronvolt, or 10^{18} eV) as expected for GZK neutrinos, were in the PeV range: 1.040 PeV and 1.140 PeV. The events were particle showers initiated by electron or tau neutrinos interacting inside the instrumented detector volume. Their light pool of roughly one hundred thousand photoelectrons extended over more than 500 m; one of them is shown in Fig. 1. With PeV energy and no trace for accompanying muons from an atmospheric shower, they are unlikely to be atmospheric in origin. It is indeed important to realize that the muon produced in the same pion or kaon decay as an atmospheric neutrino is guaranteed to reach the detector provided that the neutrino energy is sufficiently high and the zenith angle sufficiently small^{21,22}. In this case, the atmospheric neutrino provides its own self-veto. This self-veto is routinely applied to IceCube cosmic neutrino candidates that exclusively consist of isolated neutrino events.

The observation immediately suggested an analysis searching for neutrinos originating inside the detector, their well-measured energy allowing a clear separation between neutrinos of atmospheric origin and those of cosmic origin. The geometry of the veto and active signal regions was optimized to reduce the background of atmospheric muons and neutrinos to a handful of events per year whilst keeping 98% of the cosmic signal. Analysing

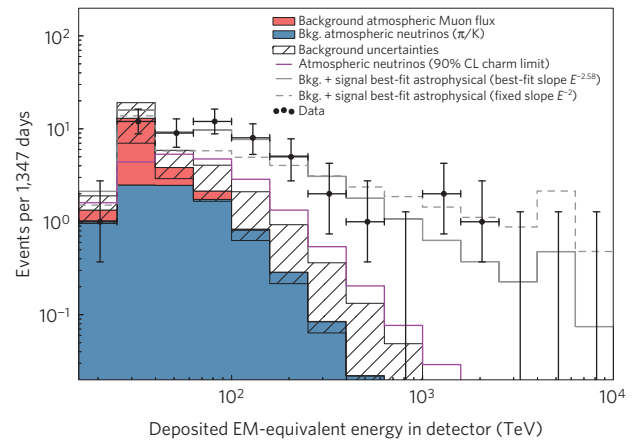


Figure 3 | Deposited energies, by neutrinos interacting inside IceCube, observed in four years of data. The hashed region shows uncertainties on the sum of all backgrounds. The atmospheric muon flux (red) and its uncertainty is computed from simulation to overcome statistical limitations in our background measurement and scaled to match the total measured background rate. The atmospheric neutrino flux is derived from previous measurements of both the π , K and charm components of the atmospheric spectrum⁶¹. Also shown are two illustrative power-law fits to the spectrum. Data measurements are shown by the black crosses with the vertical bars indicating measurement error. EM, electromagnetic; bkg., background; CL, confidence level.

the data covering the same time period as the GZK neutrino search, 28 candidate neutrino events were identified with in-detector deposited energies between 30 and 1,140 TeV. Of these, 21 are showers. The remaining seven events are muon tracks. The 28 events include the two PeV events previously revealed in the GZK neutrino search.

The energy and zenith angle dependence observed is consistent with expectations for a flux of neutrinos produced by cosmic accelerators; see Fig. 3. The flavour composition of the flux is, after corrections for the acceptances of the detector to the different flavours, consistent with $\nu_e:\nu_\mu:\nu_\tau \sim 1:1:1$, as anticipated for a flux originating in cosmic sources. It is also consistent with the flux of muon neutrinos penetrating the Earth in Fig. 2. Subsequently, two additional years of data have been analysed, doubling the statistics of the discovery publication²³. A purely atmospheric explanation can be excluded at 7σ . The four-year data set contains a total of 54 neutrino events, with deposited energies ranging from 30 to 2,000 TeV. In the fourth year, muon tracks were found that deposited ~ 500 TeV energy inside the detector, produced by PeV-energy parent neutrinos. One of them reconstructs through IceTop, IceCube's surface array, with no evidence for an air shower. Combining the absence of an air shower in IceTop with large deposited muon energy results in a high significance for astrophysical origin from a single event.

In summary, both methods for selecting cosmic neutrinos harvest about 10–15 events per year, twice that if one can tolerate a modest background. With six years of data from the completed detector, our data sample exceeds one hundred events, approaching the statistics required to identify their origin by matching arrival directions with astronomical maps²⁴.

Origin of cosmic neutrinos

Figure 4 shows in Galactic coordinates the arrival directions of cosmic neutrinos for four years of events with interaction vertices inside the detector. Shower events are labelled (+), track events (×). The colour scale indicates the value of the test statistic of an unbinned maximum likelihood test searching for anisotropies of the

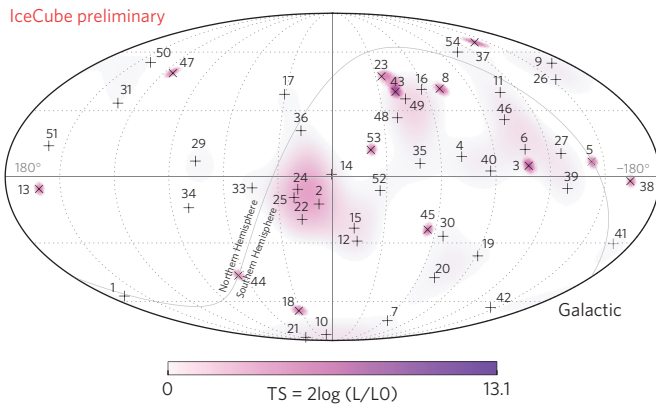


Figure 4 | Arrival directions of neutrinos in the four-year starting-event sample in Galactic coordinates. Shower-like events are shown with '+' and those containing muon tracks with 'x'. The colour scale indicates the value of the test statistic (TS) of an unbinned maximum likelihood test searching for anisotropies of the event arrival directions. Note that the track-like event 28 has been omitted following the discussion in ref. 23.

neutrino arrival directions. This measured distribution corresponds to no significant local excess in the sky; the probability of obtaining the measured anisotropy in randomized pseudo-experiments is 58%. The correlation of neutrino events with the Galactic plane is not significant. Letting the width of the plane float freely, the best fit returned a correlation for a value of $\pm 7.5^\circ$ with a post-trial chance probability of 3.3%. Neither probability increased after doubling the data from two to four years.

In short, the observed neutrino flux is consistent with an isotropic distribution of arrival directions and equal contributions of all neutrino flavours²⁵. A subdominant Galactic component of the flux cannot be excluded at this time. Interestingly, a variety of analyses^{25,26} suggest that the cosmic neutrino flux dominates the atmospheric background above an energy that may be as low as 30 TeV, with an energy spectrum that cannot be described as a single power, as was the case for the muon neutrino flux through the Earth for energies exceeding 220 TeV. This is reinforced by the fact that fitting the excess flux in different ranges of energy yields different values for the power-law exponent.

Rather than speculate on their origin, we will focus on the multimessenger connection of cosmic neutrinos to cosmic rays and gamma rays; this will turn out to be revealing. Neutrinos are produced in association with the cosmic-ray beam. Cosmic rays accelerated in regions of high magnetic fields near black holes or neutron stars inevitably interact with radiation surrounding them. In particle physics language, cosmic-ray accelerators are beam dumps. In supernova shocks, cosmic rays interact with the hydrogen in the Galactic disk, producing equal numbers of pions of all three charges that decay into pionic photons and neutrinos. These secondary fluxes should be boosted by the interaction of the cosmic rays with high-density molecular clouds that are ubiquitous in the star-forming regions where supernovae are more likely to explode. For extragalactic sources, the neutrino-producing target may be light, for instance, photons radiated by the accretion disk of an active galactic nucleus or synchrotron photons that coexist with protons in the expanding fireball producing a gamma-ray burst.

Detailed estimates indicate that, in cosmic beam dumps, roughly equal energies are injected in secondary cosmic rays (which may be the decay products of neutrons escaping the accelerator with high energy), photons, and neutrinos. Given this prediction of energy equipartition, it is remarkable that IceCube's measured energy density in cosmic neutrinos matches the total energy observed in extragalactic cosmic rays. This is especially interesting, because the

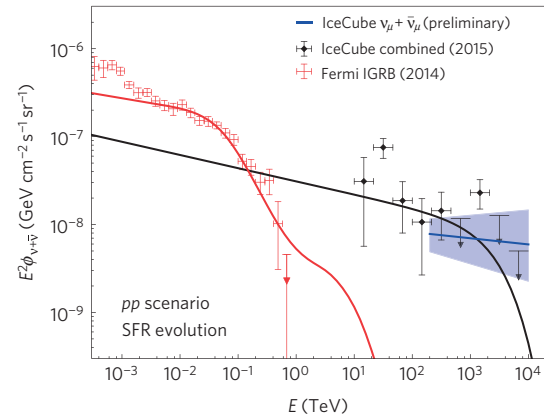


Figure 5 | Figure showing that the astrophysical neutrino flux (black line) observed by IceCube matches the corresponding cascaded gamma-ray flux (red line) observed by Fermi. We here assume that the decay products of neutral and charged pions from pp interactions are responsible for the non-thermal emission in the Universe⁶². The black data points are combined IceCube results, including the three-year 'high-energy starting event' (HESE) analysis²³ and a subsequent analysis lowering the energy threshold for events starting in the detector even further²⁶. Also shown is the best fit to the flux of high-energy muon neutrinos penetrating the Earth SFR, star formation.

parents of PeV neutrinos should have energies in the region of 10^{17} eV, well below the ankle in the spectrum at 4×10^{18} eV where traditionally the onset of the extragalactic flux has been theorized.

The spectral production rates $dN/dE dt$ of neutrinos and gamma rays are related by

$$\frac{1}{3} \sum E_\nu^2 \frac{dN_\nu}{dE_\nu dt}(E_\nu) \simeq \frac{K_\pi}{4} E_\gamma^2 \frac{dN_\gamma}{dE_\gamma dt}(E_\gamma) \quad (1)$$

Here N and E denote the number and energy of neutrinos and gamma rays and ν stand for the neutrino flavour. Note that this relation is solid, and depends only on the charged-to-neutral secondary pion ratio with $K_\pi = 1(2)$ for $\gamma(pp)$ neutrino-producing interactions. However, the production rate of gamma rays described by equation (1) is not necessarily the emission rate observed. For instance, in cosmic accelerators that efficiently produce neutrinos via $p\gamma$ interactions, the target photon field can also efficiently reduce the pionic gamma rays via pair production. Also, inverse-Compton scattering and synchrotron emission in magnetic fields will shift the emitted gamma-ray spectrum to lower energies. This is a calorimetric process that will, however, conserve the total energy of hadronic gamma rays.

It is straightforward to apply the multimessenger relation to the cosmic neutrino flux observed by IceCube. Figure 5 shows the gamma-ray flux accompanying the observed neutrino flux for two illustrative descriptions of the data that assume $K_\pi = 2$. The black and red lines show the neutrino and gamma-ray spectra after accounting for the cascading of the PeV photons in cosmic radiation backgrounds between source and observation. The black line shows an $E^{-2.15}$ neutrino spectrum with an exponential cutoff around PeV. This scenario actually matches the extragalactic isotropic diffuse gamma-ray background measured by Fermi²⁷.

This exercise indicates that the contribution of gamma rays accompanying IceCube neutrinos to Fermi's extragalactic flux is significant, suggesting a common origin of some of the sources at some level. This is intriguing, because a recent analysis indicates that blazars—active galaxies with the jet pointing at the observer—dominate the Fermi diffuse flux²⁴. Are blazars the final answer? The good news is that IceCube, by accumulating

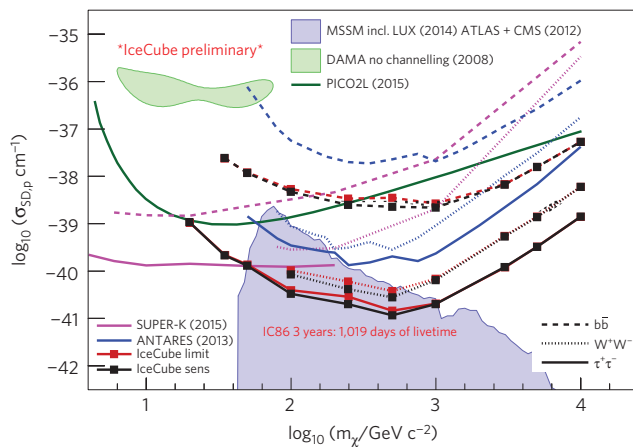


Figure 6 | Upper limits at 90% confidence level on the spin-dependent neutralino-proton cross-section assuming that the neutrinos are produced by $b\bar{b}$, $\tau\bar{\tau}$ and W^+W^- annihilation. Limits from IceCube^{41,45,46}, Super-K⁶³, ANTARES⁶⁴, Baikal⁶⁵ and Baksan⁶⁶ are shown. Full lines refer to limits on the annihilation channel and dashed lines to the $b\bar{b}$ channel. Direct search results from PICO⁶⁷ and tentative signal regions^{68–70} (grey-shaded areas) are included for comparison. The brown-shaded region indicates the allowed parameter space in MSSM supersymmetric dark matter models that are not ruled out by other experiments.

more events, can eventually identify blazars by observing multiple neutrinos from the same sources²⁴. On the other hand, a search for neutrinos from Fermi's identified blazars²⁸ has come up empty. However, these represent only 50% of the total diffuse background, allowing for the possibility that a partially different source population produces the neutrinos. This could be radio galaxies with misaligned jets. Although no definite identification of the sources of cosmic neutrinos has yet emerged, it is rather clear that a multiwavelength path to the neutrino sources looks very promising. Not unexpectedly, evidence for an association has already been claimed by some studies^{29–32}.

High-energy neutrinos and dark matter

Neutrino telescopes are powerful tools in the search for the particle nature of dark matter. By using the deepest ice with a higher density of optical sensors, IceCube's DeepCore lowers the threshold of the detector to ~ 10 GeV over a significant fraction of the detector volume; see Fig. 1. It was initially proposed as a way to enhance IceCube's capabilities for detecting lower-mass dark matter particles. It is worth noting that the AMANDA detector, the forerunner and proof of concept for IceCube, received a significant fraction of its initial funding to search for dark matter. Also, in this context, some have considered the isotropic arrival directions of cosmic neutrinos to be a clue that they originate in the Galactic halo as a result of the decay of PeV-energy dark matter particles, a speculation that at this point is perfectly consistent with observations^{33–40}.

More traditionally, IceCube searches for dark matter by looking for the annihilation of weakly interacting massive particles (WIMPs) wherever they have accumulated to a high density: in the Sun⁴¹, in the Milky Way^{42,43}, and in nearby galaxies⁴⁴. For instance, WIMPs are swept up by the Sun as the solar system moves about the galactic halo. Although interacting weakly, they will occasionally scatter elastically with nuclei in the Sun and lose sufficient momentum to become gravitationally bound. Over the lifetime of the Sun, WIMPs may accumulate to a density where equilibrium is established between their capture and annihilation. The annihilation of these WIMPs to final states which can decay to neutrinos represents an indirect signature of halo dark matter. This WIMP annihilation signal is revealed by the neutrinos that escape the Sun with minimal

absorption. The neutrinos are, for instance, the decay products of heavy quarks and weak bosons resulting from the annihilation of WIMPs into $\chi\chi \rightarrow \tau\bar{\tau}$, $b\bar{b}$, or W^+W^- . Neutrino telescopes are sensitive to such neutrinos because of their relatively high energy, above 20 GeV at this point, reflecting the mass of the decaying WIMP.

The beauty of the indirect detection technique using neutrinos originating from the Sun is that the astrophysics of the problem is understood. The source in the Sun has built up over solar time, sampling the dark matter throughout the galaxy; therefore, any possible structure in the halo has been averaged out. Given a WIMP mass and properties, one can unambiguously predict the signal in a neutrino telescope; if not observed, the model is ruled out. This is in contrast to other indirect searches whose sensitivity depends critically on the structure of halo dark matter; observation requires cuspy structure near the galactic centre or clustering on appropriate scales elsewhere. Observation not only necessitates appropriate WIMP properties but also favourable astrophysical circumstances.

IceCube has established world-leading limits on WIMPs with significant spin-dependent interactions with protons, because they result in strong concentrations inside the Sun, a nearby and readily identifiable source^{45,46}. An excess of neutrinos, of GeV or higher energy, over the atmospheric neutrino background in the direction of the Sun is the signature of dark matter. There is no alternative astrophysical explanation of such a signal, which represents a smoking gun for dark matter particles. In most WIMP scenarios, the cross-sections for WIMP capture and for WIMP annihilation are large enough so that an equilibrium between capture and annihilation would have been achieved within the age of the solar system⁴⁷. In this case, limits on neutrinos from the Sun can be expressed in terms of the capture cross-section. If equilibrium is not reached, weaker limits can still be derived.

The current IceCube limits^{41,45,46} are shown in Fig. 6. These are derived from three years of muon neutrino observations; no excess of neutrinos over the atmospheric flux is found in the direction of the sun. By including events that start inside DeepCore, the mass range for the WIMP search could be extended down to 20 GeV, which overlaps some of the allowed region from the DAMA experiment⁴⁸. Since the exact branching ratios of WIMP annihilation into different channels is model-dependent, experiments usually choose two annihilation channels that give extreme neutrino spectra to show their results. Annihilation into $b\bar{b}$ is chosen as a representative case producing a soft neutrino spectrum, and annihilation into W^+W^- or $\tau\bar{\tau}$ as a hard spectrum. Assuming a 100% branching ratio to each of these channels brackets the expected neutrino spectrum for any model with branching to more channels. IceCube and Super-K reach bounds at the $10^{-40} - 10^{-41} \text{ cm}^2$ level, covering the WIMP mass range, between the two experiments, from a few GeV to 100 TeV. Because of the A^2 coherence factor for scattering on heavy nuclei with atomic number A , the direct detection experiments have an advantage over IceCube for the case of spin-independent interactions. They thus achieve superior limits to IceCube.

Future developments

By building IceCube, it was possible to map the optical properties of natural ice over large distances. We made the surprising discovery that the absorption length of the Čerenkov light to which the photomultipliers are sensitive exceeds 100 m. In fact, in the lower half of the detector it exceeds 200 m. The absorption length dictates the distance by which one can space the strings of sensors without spoiling the uniformity of the detector. Current modelling indicates that spacings of 250 m, possibly larger, are acceptable. One can therefore instrument a ten-times-larger volume of ice with the same number of strings used to build IceCube. The design of such a next-generation instrument using superior light sensors and data

acquisition electronics is in progress⁴⁹. IceCube-Gen2 will turn a fascinating discovery into an instrument for neutrino astronomy.

Received 18 March 2016; accepted 8 June 2016;
published online 4 July 2016

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Acknowledgements

Discussion with collaborators inside and outside the IceCube Collaboration, too many to be listed, have greatly shaped this presentation. Thanks. This research was supported in part by the US National Science Foundation under Grant Nos ANT-0937462 and PHY-1306958 and by the University of Wisconsin Research Committee with funds granted by the Wisconsin Alumni Research Foundation.

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Competing financial interests

The author declares no competing financial interests.